LOCALIZATION FUNCTORS PRESERVE SCHEMES AND ALGEBRAIC SPACES UNDER A CHANGE OF BASE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The statement is a six-part assertion; the two enumerate blocks are part of the counted statement span and are preserved verbatim.

Editorial note. The eight references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: sources/stacks-spaces/retained-preamble.tex, sliced mechanically from the chapter's own preamble.tex.

Allowed standard prerequisites (editorial).

- Definition of the localization functors j_!, j_*, j^{-1}. Sites, Lemma sites-lemma-relocalize, used at proof lines 2982-2986. Makes the localization material available for j.
- The coproduct formula for j_!. Sites, Lemma sites-lemma-describe-j-shriek-good-site, used at proof lines 2987-2988. Gives assertion (1).
- j_! commutes with fibre products and equalizers. Sites, Lemma sites-lemma-j-shriek-commutes-equalizers-fibre-products, used at proof lines 2990-2992. Transports the equivalence relation.
- j_! of a representable sheaf is representable. Sites, Lemma sites-lemma-describe-j-shriek-representable, used at proof lines 2993-2995. Gives assertion (2).
- Presentation of an algebraic space and the presentation theorem. Lemma lemma-space-presentation (1387-1407) and Theorem theorem-presentation, used at proof lines 2996-3009 and 3018-3024. Recognizes algebraic spaces from etale equivalence relations.

Source reference. Original lemma 0262: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2835–2835. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 03HT: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4579–4579. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 03HU: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4580–4580. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Date: stacks-project:spaces.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 04BB: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 5646–5646. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 03CE: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4381–4381. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 03EH: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4456–4456. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original section 00XZ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 1237–1237. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original theorem 02WW: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3819–3819. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Suppose given a big site Sch_{fppf}. Let $g : S \to S'$ be morphism of Sch_{fppf}. Let $j : (Sch/S)_{fppf} \to (Sch/S')_{fppf}$ be the corresponding localization functor. Let F be a sheaf of sets on $(Sch/S)_{fppf}$. Then*

1. *for a scheme T' over S' we have $j_!F(T'/S') = \coprod_{\varphi : T' \to S} F(T' \xrightarrow{\varphi} S)$,*
2. *if F is representable by a scheme $X \in \mathrm{Ob}((Sch/S)_{fppf})$, then $j_!F$ is representable by $j(X)$ which is X viewed as a scheme over S', and*
3. *if F is an algebraic space over S, then $j_!F$ is an algebraic space over S', and if $F = U/R$ is a presentation, then $j_!F = j(U)/j(R)$ is a presentation.*

Let F' be a sheaf of sets on $(Sch/S')_{fppf}$. Then

4. *for a scheme T over S we have $j^{-1}F'(T/S) = F'(T/S')$,*
5. *if F' is representable by a scheme $X' \in \mathrm{Ob}((Sch/S')_{fppf})$, then $j^{-1}F'$ is representable, namely by $X'_S = S \times_{S'} X'$, and*
6. *if F' is an algebraic space, then $j^{-1}F'$ is an algebraic space, and if $F' = U'/R'$ is a presentation, then $j^{-1}F' = U'_S/R'_S$ is a presentation.*

Proof. The functors $j_!$, j_* and j^{-1} are defined in Sites, Lemma 03EH where it is also shown that $j = j_{S/S'}$ is the localization of $(Sch/S')_{fppf}$ at the object S/S'. Hence all of the material on localization functors is available for j. The formula in (1) is Sites, Lemma 03HT. By definition $j_!$ is the left adjoint to restriction j^{-1}, hence $j_!$ is right exact. By Sites, Lemma 04BB it also commutes with fibre products and equalizers. By Sites, Lemma 03HU we see that $j_!h_X = h_{j(X)}$ hence (2) holds. If F is an algebraic space over S, then we can write $F = U/R$ (Lemma 0262) and we get

$$j_!F = j(U)/j(R)$$

because $j_!$ being right exact commutes with coequalizers, and moreover $j(R) = j(U) \times_{j_!F} j(U)$ as $j_!$ commutes with fibre products. Since the morphisms $j(s), j(t) : j(R) \to j(U)$ are simply the morphisms $s, t : R \to U$ (but viewed as morphisms of schemes over S'), they are still étale. Thus $(j(U), j(R), s, t)$ is an étale equivalence relation. Hence by Theorem 02WW we conclude that $j_!F$ is an algebraic space.

Proof of (4), (5), and (6). The description of j^{-1} is in Sites, Section 00XZ. The restriction of the representable sheaf associated to X'/S' is the representable sheaf associated to $X'_S = S \times_{S'} Y'$ by Sites, Lemma 03CE. The restriction functor j^{-1} is exact, hence $j^{-1}F' = U'_S/R'_S$. Again by exactness the sheaf R'_S is still an equivalence relation on U'_S. Finally the two maps $R'_S \to U'_S$ are étale as base changes of the étale morphisms $R' \to U'$. Hence $j^{-1}F' = U'_S/R'_S$ is an algebraic space by Theorem 02WW and we win. □